COURSE 5 · ASTRONOMY

Astronomy

Printable workbook with worked solutions

64 problems across 8 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 64 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook in the Second Pass app, which checks answers instantly. The app also has a quiz for every module and course, and spaced-repetition flashcards; a printable set of the flashcards comes with each course.

> **A good routine for each module**
>
> Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then take the module quiz in the app and redo anything you got wrong without looking. Review the module's flashcards for a few minutes each day after that.

Module 5.1

The sky and the history of astronomy

5.1.1 ●○○

What causes Earth's seasons?

A. changing distance from the Sun

B. the tilt of Earth's axis

C. the Moon

D. changes in the Sun's output

5.1.2 ●○○

Why does the Moon show phases?

A. Earth's shadow

B. we see different amounts of its sunlit half

C. it changes shape

D. clouds

5.1.3 ●●○

Brisbane is at latitude 27.5° S. What is the Sun's noon altitude at the December (summer) solstice, in degrees?

Answer: _____ °

5.1.4 ●●○

And at the June (winter) solstice?

Answer: _____ °

5.1.5 ●●○

In which month is Earth closest to the Sun?

A. January

B. April

C. July

D. October

5.1.6 ●●○

Who first showed planetary orbits are ellipses?

A. Copernicus

B. Ptolemy

C. Kepler

D. Galileo

5.1.7 ●●○

Galileo's observation that Venus shows a full set of phases supported

A. the Earth-centred model

B. the Sun-centred model

C. flat Earth

D. nothing

5.1.8 ●●●

Why can't people in Europe see the Southern Cross?

Module 5.2

Gravity and orbits

5.2.1 ●○○

A planet orbits at 4 AU from the Sun. What is its period in years?

Answer: _____ years

5.2.2 ●●○

If the distance between two masses triples, by what factor does the gravitational force change? Give a decimal.

Answer: _____

5.2.3 ●●○

Mars has mass 6.42×10^{23} kg and radius 3.39×10^{6} m. What is its surface gravity?

Answer: _____ m/s^2

5.2.4 ●●○

The ISS orbits at $r = 6.78 \times 10^{6}$ m from Earth's centre. Earth's $GM = 3.99 \times 10^{14}$ m^3/s^2. What is its orbital speed?

Answer: _____ m/s

5.2.5 ●●●

The Moon orbits Earth at 3.84×10^{8} m with a period of 27.3 days. Estimate Earth's mass.

Answer: _____ kg

5.2.6 ●●○

Astronauts in the ISS float because

A. there is no gravity there

B. they are in free fall with the station

C. the station blocks gravity

D. they are too far from Earth

5.2.7 ●●○

What is the escape velocity from the Moon? ($GM = 4.9 \times 10^{12}$ m^3/s^2, $R = 1.74 \times 10^6$ m)

Answer: _____ m/s

5.2.8 ●●○

By Kepler's second law, a comet moves fastest

A. far from the Sun

B. closest to the Sun

C. at a constant speed

D. when crossing Earth's orbit only

Module 5.3

Light as messenger

5.3.1 ●○○

A star is moved to three times its distance. By what factor does its apparent brightness change? Give a decimal.

Answer: _____

5.3.2 ●○○

A star has a parallax of 0.25 arcseconds. How far away is it, in parsecs?

Answer: _____ pc

5.3.3 ●●○

Convert 4 parsecs to light-years.

Answer: _____ ly

5.3.4 ●●○

The Sun's surface is about 5800 K. What is its peak wavelength in nanometres?

Answer: _____ nm

5.3.5 ●●○

A star's peak wavelength is 290 nm. What is its surface temperature?

Answer: _____ K

5.3.6 ●●○

A hydrogen line normally at 656.3 nm is observed at 656.9 nm. How fast is the star receding, in km/s? ($c = 3.0 \times 10^5$ km/s)

Answer: _____ km/s

5.3.7 ●●○

Why are X-ray telescopes placed in space?

A. to be closer to stars

B. Earth's atmosphere absorbs X-rays

C. to avoid light pollution only

D. X-rays need vacuum to exist

5.3.8 ●●●

How did astronomers know helium existed before it was found on Earth?

Module 5.4

The Solar System

5.4.1 ●○○

What powers the Sun?

A. burning gas

B. nuclear fission

C. nuclear fusion

D. gravity only

5.4.2 ●○○

Jupiter is 5.2 AU from the Sun. How many kilometres is that? (1 AU = 1.5×10^8 km)

Answer: _____ km

5.4.3 ●●○

How long does sunlight take to reach Neptune, 30 AU away, in hours? (Earth: 500 s)

Answer: _____ h

5.4.4 ●●○

Why are the inner planets rocky and the outer ones gas giants?

A. the inner ones are older

B. near the Sun it was too hot for ices to condense

C. the Sun ate their gas later

D. chance

5.4.5 ●●○

Why do the planets orbit in nearly the same plane?

A. coincidence

B. they formed from a flattened spinning disc

C. the Sun's magnetism

D. they were captured

5.4.6 ●●○

A Jupiter-sized planet (radius 0.1 × the star's) transits its star. By what fraction does the starlight dim? Give a decimal.

Answer: _____

5.4.7 ●●●

An Earth-sized planet (radius 0.009 × the Sun's) transits a Sun-like star. What is the fractional dimming?

Answer: _____

5.4.8 ●●●

Why is Venus hotter than Mercury even though it is further from the Sun?

Module 5.5

Stars

5.5.1 ●○○

Two stars are the same size, but one is twice as hot. How many times more luminous is it?

Answer: _____ ×

5.5.2 ●●○

A star has the same temperature as the Sun but 10 times its radius. How many times more luminous is it?

Answer: _____ ×

5.5.3 ●●○

On the HR diagram, a hot but very faint star is most likely a

A. red giant

B. main-sequence star

C. white dwarf

D. supergiant

5.5.4 ●●○

Using $t \approx 10^{10} \times M^{(-2.5)}$ years, estimate the lifetime of a star of 4 solar masses.

Answer: _____ years

5.5.5 ●●○

What keeps a stable star from collapsing?

A. rotation

B. pressure from hot gas and radiation

C. magnetism

D. nothing; they are collapsing slowly

5.5.6 ●●○

In hydrogen fusion about 0.7% of mass becomes energy. How much energy comes from fusing 1 kg of hydrogen? ($c = 3 \times 10^8$ m/s)

Answer: _____ J

5.5.7 ●●○

Where do stars spend most of their lives?

A. as giants

B. on the main sequence

C. as white dwarfs

D. as nebulae

5.5.8 ●●●

Massive stars have much more fuel than the Sun. Why do they live much shorter lives?

Module 5.6

Stellar death

5.6.1 ●○○

What will the Sun become at the end of its life?

A. a black hole

B. a neutron star

C. a white dwarf

D. a supernova

5.6.2 ●○○

Where were most of the heavy elements in your body made?

A. the Big Bang

B. inside stars and supernovae

C. on Earth

D. in the Sun

5.6.3 ●●○

What is the Schwarzschild radius of a 10 solar-mass black hole, in km? ($M_\odot = 2.0 \times 10^{30}$ kg, $G = 6.67 \times 10^{-11}$, $c = 3.0 \times 10^{8}$)

Answer: _____ km

5.6.4 ●●○

A white dwarf has mass 1.2×10^{30} kg and radius 6.0×10^{6} m. What is its density, in kg/m^3?

Answer: _____ kg/m^3

5.6.5 ●●○

Why does fusion stop at iron in massive stars?

A. iron is too rare

B. fusing iron absorbs rather than releases energy

C. iron is not a gas

D. the star is too cold

5.6.6 ●●○

Pulsars are

A. pulsating red giants

B. rotating neutron stars

C. black holes

D. planets

5.6.7 ●●○

What is the maximum mass of a white dwarf?

A. about 0.5 solar masses

B. about 1.4 solar masses

C. about 10 solar masses

D. no limit

5.6.8 ●●●

If the Sun were replaced by a black hole of the same mass, what would happen to Earth's orbit, and why?

Module 5.7

Galaxies

5.7.1 ●○○

What type of galaxy is the Milky Way?

A. elliptical

B. irregular

C. barred spiral

D. lenticular only

5.7.2 ●●○

The Sun orbits the galactic centre at 230 km/s at a radius of 2.5×10^{20} m. How long is one orbit, in millions of years? (1 year = 3.15×10^{7} s)

Answer: _____ million years

5.7.3 ●●○

Using $M = v^2 r / G$ with the numbers above, estimate the mass inside the Sun's orbit, in kg.

Answer: _____ kg

5.7.4 ●●○

Flat galaxy rotation curves are evidence for

A. black holes

B. dark matter

C. dark energy

D. spiral arms

5.7.5 ●●○

Andromeda is 2.5 million light-years away. When we look at it, how long ago did that light leave, in years?

Answer: _____ years

5.7.6 ●●○

Elliptical galaxies typically contain

A. lots of gas and young stars

B. mostly old stars and little gas

C. no stars

D. only black holes

5.7.7 ●●○

How was the mass of the black hole at the Milky Way's centre measured?

A. by its light

B. from the orbits of nearby stars

C. by sound

D. by its colour

5.7.8 ●●●

Why did astronomers expect stars far from a galaxy's centre to orbit more slowly, and what did they find?

Module 5.8

Cosmology

5.8.1 ●○○

A galaxy is 100 Mpc away. Using H_0 = 70 km/s/Mpc, how fast is it receding?

Answer: _____ km/s

5.8.2 ●●○

A galaxy recedes at 14 000 km/s. How far away is it?

Answer: _____ Mpc

5.8.3 ●●○

Estimate the age of the universe as $1/H_0$. H_0 = 70 km/s/Mpc, 1 Mpc = 3.09×10^{19} km, 1 year = 3.15×10^{7} s. Give billions of years.

Answer: _____ billion years

5.8.4 ●●○

A line emitted at 500 nm is observed at 600 nm. What is the redshift z?

Answer: _____

5.8.5 ●●○

The cosmic microwave background is

A. light from the first stars

B. light released when atoms first formed, now stretched to microwaves

C. radio noise from Earth

D. light from black holes

5.8.6 ●●○

Which is NOT one of the main pillars of evidence for the Big Bang?

A. expansion of the universe

B. cosmic microwave background

C. light-element abundances

D. the existence of black holes

5.8.7 ●●○

Roughly what fraction of the universe is ordinary matter?

A. 5%

B. 27%

C. 68%

D. 100%

5.8.8 ●●●

If all galaxies are moving away from us, does that mean we are at the centre of the universe? Explain.

Capstone

Characterising a star with a planet

A Sun-like star 15 parsecs away hosts a transiting planet. Astronomers measure its spectrum, brightness and the planet's orbit. Use the tools of this course to find out about both. Take $G = 6.67 \times 10^{-11}$, the Sun's mass 2.0×10^{30} kg, 1 AU = 1.5×10^{11} m.

Part 1 ●●○

What is the star's parallax angle, in arcseconds?

Answer: _____ arcsec

Part 2 ●●○

Its spectrum peaks at 500 nm. What is its surface temperature?

Answer: _____ K

Part 3 ●●○

The planet orbits every 365 days, and the star has the Sun's mass. How far is the planet from the star, in AU?

Answer: _____ AU

Part 4 ●●○

During transit, the starlight dims by 0.01%. What is the planet's radius as a fraction of the star's?

Answer: _____

Part 5 ●●●

What is the planet's orbital speed, in km/s?

Answer: _____ km/s

Part 6 ●●●

Is this planet a good candidate for life? Give two reasons for and one reason for caution.

Answers

Worked solutions

Module 5.1 · The sky and the history of astronomy

5.1.1 ANSWER B. the tilt of Earth's axis

1 Tilt changes how directly sunlight strikes each hemisphere.

5.1.2 ANSWER B. we see different amounts of its sunlit half

1 Half is always lit; our viewing angle changes.

5.1.3 ANSWER 85.9 °

1 h = 90 − 27.5 + 23.4.

2 ≈ 85.9°, almost overhead.

5.1.4 ANSWER 39.1 °

1 h = 90 − 27.5 − 23.4 ≈ 39.1°.

5.1.5 ANSWER A. January

1 Perihelion is around 3–4 January.

5.1.6 ANSWER C. Kepler

1 Kepler, using Tycho's data, 1609.

5.1.7 ANSWER B. the Sun-centred model

1 Full phases require Venus to orbit the Sun.

5.1.8 ANSWER The Southern Cross lies close to the south celestial pole. From northern latitudes, that part of the celestial sphere stays permanently below the horizon as Earth rotates, so it never rises. You need to be roughly south of about 25° N to see it.

1 Think about the celestial sphere and your latitude.

Module 5.2 · Gravity and orbits

5.2.1 ANSWER 8 years

1 $T^2 = 4^3 = 64$.

5.2.2 ANSWER 0.1111

1 Inverse square: $1/3^2 = 1/9 \approx 0.111$.

5.2.3 ANSWER 3.726 m/s²

1 g = GM/R².

2 ≈ 3.7 m/s².

5.2.4 ANSWER 7671 m/s

1 v = √(GM/r).

2 ≈ 7670 m/s.

5.2.5 ANSWER 6.02 × 10²⁴ kg

1 M = 4π²r³/(GT²).

2 T = 2.36 × 10⁶ s; M ≈ 6.0 × 10²⁴ kg.

5.2.6 ANSWER B. they are in free fall with the station

1 Gravity there is about 90% of surface strength.

5.2.7 ANSWER 2373 m/s

1 v = √(2GM/R).

2 ≈ 2370 m/s.

5.2.8 ANSWER B. closest to the Sun

1 Equal areas in equal times.

Module 5.3 · Light as messenger

5.3.1 ANSWER 0.1111

1 Inverse square: 1/9.

5.3.2 ANSWER 4 pc

1 d = 1/p = 1/0.25.

5.3.3 ANSWER 13.04 ly

1 4 × 3.26 ≈ 13 ly.

5.3.4 ANSWER 500 nm

1 λ = 2.90 × 10⁻³ / 5800 = 5.0 × 10⁻⁷ m.

2 = 500 nm, green-blue.

5.3.5 ANSWER 1e+04 K

1 T = 2.90 × 10⁻³ / 2.90 × 10⁻⁷ = 10 000 K.

5.3.6 ANSWER 274.3 km/s

1 $\Delta\lambda/\lambda = 0.6/656.3$.

2 $v = 9.14 \times 10^{-4} \times 3 \times 10^{5} \approx 274$ km/s.

5.3.7 ANSWER B. Earth's atmosphere absorbs X-rays

1 The atmosphere is opaque to X-rays.

5.3.8 ANSWER In 1868, during a solar eclipse, a bright spectral line was seen in the Sun's light at a wavelength that matched no known element. Since every element has a unique set of lines, it had to be a new element, named after the Greek Helios. It was isolated on Earth in 1895.

1 Spectral lines are fingerprints.

Module 5.4 · The Solar System

5.4.1 ANSWER C. nuclear fusion

1 Hydrogen fuses into helium in the core.

5.4.2 ANSWER 7.8×10^{8} km

1 $5.2 \times 1.5 \times 10^{8} = 7.8 \times 10^{8}$ km.

5.4.3 ANSWER 4.167 h

1 30×500 s = 15 000 s.

2 $\div 3600 \approx 4.2$ h.

5.4.4 ANSWER B. near the Sun it was too hot for ices to condense

1 The frost line.

5.4.5 ANSWER B. they formed from a flattened spinning disc

1 Conservation of angular momentum flattened the cloud.

5.4.6 ANSWER 0.01

1 $(0.1)^2 = 0.01$, a 1% dip.

5.4.7 ANSWER 8.1×10^{-5}

1 $(0.009)^2 \approx 8.1 \times 10^{-5}$: less than one part in ten thousand.

5.4.8 ANSWER Venus has a very thick carbon dioxide atmosphere. Sunlight warms the surface, which radiates infrared; the CO_2 traps that infrared (a runaway greenhouse effect), raising the surface to about 465 °C. Mercury has almost no atmosphere, so its heat escapes and its night side gets very cold.

1 What is different about Venus's atmosphere?

Module 5.5 · Stars

5.5.1 ANSWER 16 ×

1 $L \propto T^4$.

2 $2^4 = 16$.

5.5.2 ANSWER 100 ×

1 $L \propto R^2$ at fixed T.

2 $10^2 = 100$.

5.5.3 ANSWER C. white dwarf

1 Hot and faint means small.

5.5.4 ANSWER 3.12×10^8 years

1 4^2.5 = 32.

2 $10^{10} \div 32 \approx 3.1 \times 10^8$ years.

5.5.5 ANSWER B. pressure from hot gas and radiation

1 Pressure balances gravity.

5.5.6 ANSWER 6.3×10^{14} J

1 $\Delta m = 0.007$ kg.

2 $E = 0.007 \times 9 \times 10^{16} = 6.3 \times 10^{14}$ J.

5.5.7 ANSWER B. on the main sequence

1 Fusing hydrogen in the core.

5.5.8 ANSWER Luminosity rises far faster than mass, roughly as M^3.5. A star ten times the Sun's mass has ten times the fuel but burns it thousands of times faster, so it runs out in a fraction of the time: lifetime goes roughly as $M/L \propto$ M^−2.5.

1 Compare how fuel and burning rate scale with mass.

Module 5.6 · Stellar death

5.6.1 ANSWER C. a white dwarf

1 It is too low in mass for a supernova.

5.6.2 ANSWER B. inside stars and supernovae

1 Stellar nucleosynthesis.

5.6.3 ANSWER 29.64 km

1 $R = 2GM/c^2 = 2 \times 6.67 \times 10^{-11} \times 2 \times 10^{31} \div 9 \times 10^{16}$.

2 $\approx 29\,600$ m ≈ 30 km.

5.6.4 ANSWER 1.33×10^9 kg/m^3

1 $V = (4/3)\pi(6 \times 10^6)^3 \approx 9.05 \times 10^{20}$ m^3.

2 $\rho \approx 1.3 \times 10^9$ kg/m^3.

5.6.5 ANSWER B. fusing iron absorbs rather than releases energy

1 Iron has the most tightly bound nucleus.

5.6.6 ANSWER B. rotating neutron stars

1 Discovered by Jocelyn Bell Burnell, 1967.

5.6.7 ANSWER B. about 1.4 solar masses

1 The Chandrasekhar limit.

5.6.8 ANSWER Nothing would change in the orbit. Gravity outside a spherical mass depends only on the mass and distance ($F = GMm/r^2$), not on how compact the object is. Earth would keep orbiting at the same distance and speed, though it would be dark and cold.

1 What does Newton's law depend on?

Module 5.7 · Galaxies

5.7.1 ANSWER C. barred spiral

1 A barred spiral.

5.7.2 ANSWER 216.8 million years

1 $T = 2\pi r/v = 2\pi \times 2.5 \times 10^{20} \div 2.3 \times 10^5 \approx 6.8 \times 10^{15}$ s.

2 $\div 3.15 \times 10^7 \approx 2.2 \times 10^8$ years ≈ 220 million.

5.7.3 ANSWER 1.98×10^{41} kg

1 $(2.3 \times 10^5)^2 \times 2.5 \times 10^{20} \div 6.67 \times 10^{-11}$.

2 $\approx 2.0 \times 10^{41}$ kg, about 10^{11} solar masses.

5.7.4 ANSWER B. dark matter

1 Extra unseen mass in the halo.

5.7.5 ANSWER 2.5×10^6 years

1 One light-year per year of travel.

5.7.6 ANSWER B. mostly old stars and little gas

1 They are red and dead, mostly.

5.7.7 ANSWER B. from the orbits of nearby stars

1 Kepler's third law applied to stars orbiting it.

5.7.8 ANSWER If most of the mass were concentrated in the bright centre, orbital speed far out would fall as $v = \sqrt{(GM/r)}$, like the outer planets. Instead, speeds stay roughly constant at large radius, which means the enclosed mass keeps growing with distance even where little light is seen. This implies a large halo of dark matter.

1 Compare with Kepler's laws in the Solar System.

Module 5.8 · Cosmology

5.8.1 ANSWER 7000 km/s

1 $v = H_0 d = 70 \times 100$.

5.8.2 ANSWER 200 Mpc

1 $d = v/H_0 = 14\,000/70$.

5.8.3 ANSWER 14.01 billion years

1 $1/H_0 = 3.09 \times 10^{19}$ km $\div$ 70 km/s $= 4.41 \times 10^{17}$ s.

2 $\div\ 3.15 \times 10^7 \approx 1.4 \times 10^{10}$ years.

5.8.4 ANSWER 0.2

1 $z = (600 - 500)/500$.

5.8.5 ANSWER B. light released when atoms first formed, now stretched to microwaves

1 About 380 000 years after the Big Bang.

5.8.6 ANSWER D. the existence of black holes

1 Black holes are not evidence for the Big Bang.

5.8.7 ANSWER A. 5%

1 The rest is dark matter and dark energy.

5.8.8 ANSWER No. In a uniformly expanding universe, every observer sees every other galaxy receding, with speed proportional to distance. Like dots on an inflating balloon, no dot is special. The expansion is of space itself, with no centre and no edge.

1 Imagine standing on a different galaxy.

Capstone · Characterising a star with a planet

Part 1 ANSWER 0.06667 arcsec

1 p = 1/d = 1/15 ≈ 0.067″.

Part 2 ANSWER 5800 K

1 $T = 2.90 \times 10^{-3} / 5.0 \times 10^{-7} = 5800$ K.

Part 3 ANSWER 1 AU

1 $T^2 = a^3$ in solar units.

2 T = 1 year, so a = 1 AU.

Part 4 ANSWER 0.01

1 $\Delta F/F = (Rp/Rs)^2 = 10^{-4}$.

2 Rp/Rs = 0.01.

Part 5 ANSWER 29.82 km/s

1 $v = \sqrt{(GM/r)} = \sqrt{(6.67 \times 10^{-11} \times 2 \times 10^{30} \div 1.5 \times 10^{11})}$.

2 ≈ 29.8 km/s.

Part 6 ANSWER For: it orbits a stable Sun-like star at about 1 AU, so it may receive Earth-like energy and could have liquid water; its radius (about 1% of a Sun-like star, so roughly Earth-sized) suggests a rocky world. Caution: we know nothing yet about its atmosphere, mass or composition; follow-up spectroscopy during transit would be needed to look for water or biosignature gases.

1 Think about distance from the star, size, and what we have not measured.